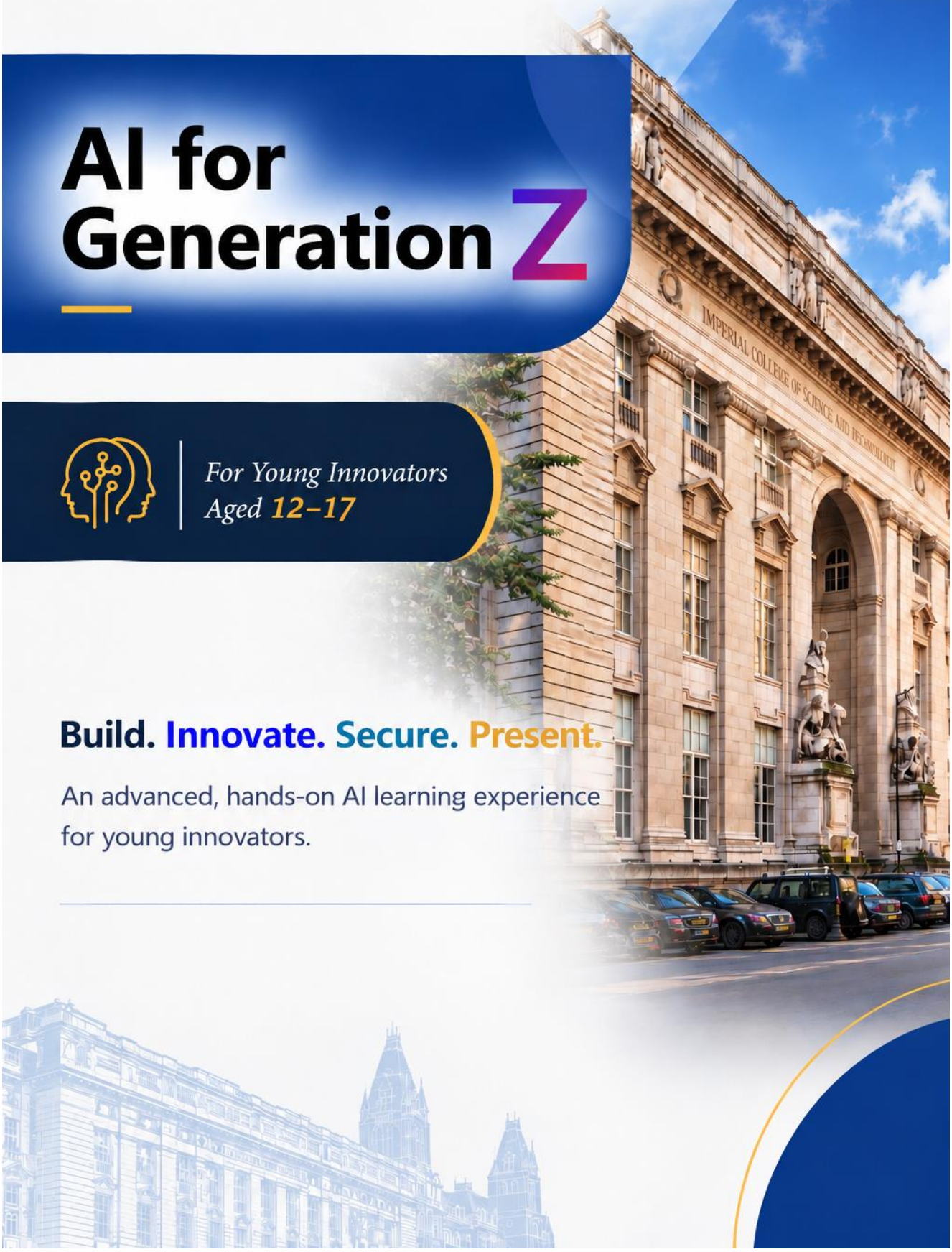
In partnership

ET THE ECONOMIC TIMES × IMPERIAL

AI for Generation Z

For Young Innovators Aged 12–17



Build. Innovate. Secure. Present.

An advanced, hands-on AI learning experience for young innovators.

01 Programme overview

Five days. One working AI product. Built, secured and presented by students.

AI Academy for the Next Generation is a five-day, hands-on applied AI programme from The Economic Times (ET), proposed in partnership with Imperial College London. It takes students from using AI tools to building, securing and presenting a working AI product.

5	40	4	6	1:10
DAYS	HOURS	PILLARS	PER TEAM	STAFF RATIO
in person	of applied learning	one build-to-demo arc	rotating daily roles	speaker + GTA per 10

DESIGNED FOR STUDENTS AGED 15–18 ·
prior experience with AI tools, including ET "AI for Students" bootcamp graduates

What it is

Four pillars over five days: Build, Innovate, Secure, Present. Every house team ends with a deployed, tested and security-reviewed AI product and a live demonstration.

Who it is for

Students aged 15–18 who already know AI tools and want to learn how AI systems are built and trusted, including graduates of ET's "AI for Students" bootcamp.

Why it stands out

It teaches responsible AI as a build skill: students red-team their own work, document risks and explain limits to a non-technical audience.

What students take away

A working AI product, a security audit log and release checklist, a reflection essay, a portfolio package and a certificate.

02 Why this programme is needed

From using AI to understanding it

Students can now use AI tools easily, but few learn how AI systems are built, tested and kept safe. This programme closes that gap with one week of applied, supervised practice.

THE GAP	WHAT STUDENTS TYPICALLY HAVE	WHAT THE PROGRAMME ADDS
Tool use without understanding	Fluency with prompting and no-code tools	How rules, machine learning, generative models and agents differ, and when each fails
Prototypes that never ship	Demos that run on one laptop	Deployed products with a CI pipeline, an evaluation set and observability
Blind trust in AI output	Little practice checking answers	Red-teaming for hallucination, bias, prompt injection and privacy leaks, then fixing what is found
Weak communication of technical work	Few chances to present to experts	Live demos, Q&A with external judges and a client-style written report
No evidence of capability	Course certificates only	A portfolio of code, demo video and reflection that applications can cite

A READY PIPELINE OF STUDENTS

The programme builds directly on ET's "AI for Students" bootcamp: 20+ tools, live expert-led sessions, hands-on projects and certification. ET already has a pipeline of students ready for the next step.

03 Programme at a glance

Everything about the week on one page

PROGRAMME	AI Academy for the Next Generation: Build · Innovate · Secure · Present
FORMAT	In person, five consecutive days, 09:00 to 17:00 (40 hours)
AUDIENCE	**Students aged 15–18 with prior AI tool experience**
ENTRY ROUTE	ET "AI for Students" bootcamp, or equivalent experience
TEAMS	Four houses of six students, with daily role rotation (24 students in the ET blueprint; final cohort size to be confirmed)
STAFFING	One speaker plus one graduate teaching assistant (GTA) per 10 students during project sessions
OUTPUT PER TEAM	One deployed, security-reviewed AI product, a written client report and a live demo
RECOGNITION	Individual certificate and feedback for every student; an Imperial-branded certificate of participation is proposed

LEARNING OUTCOMES

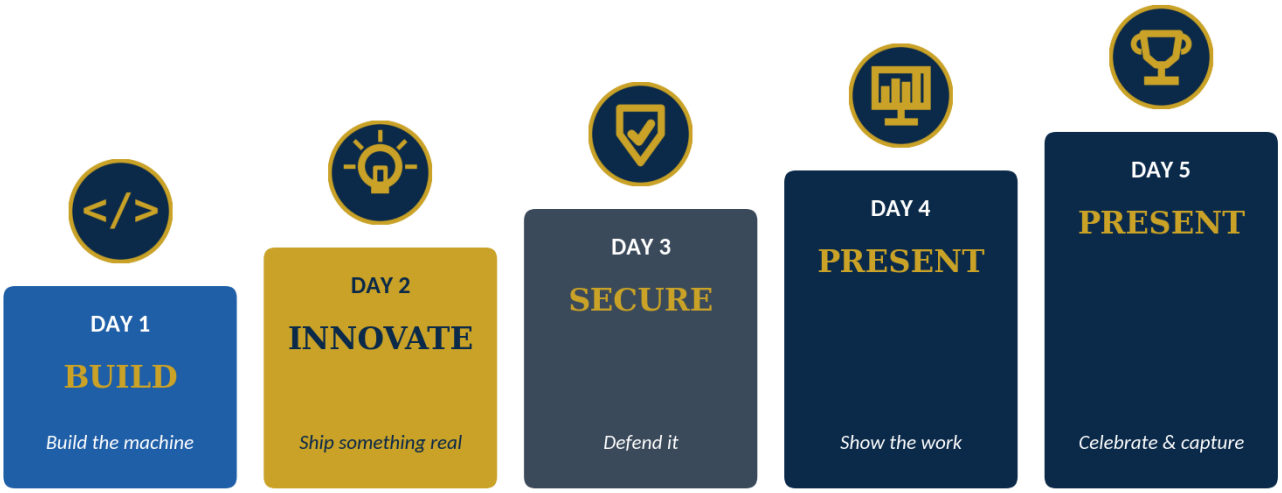
By day five, every student can:

BUILD	an agentic AI system with tool use, memory and an evaluation set, and deploy it.
INNOVATE	through a design sprint that tests an idea with real users.
SECURE	a product by attacking it, fixing the findings and documenting risk.
PRESENT	a working solution, answer expert questions and reflect on what to rebuild.
COLLABORATE	in cross-functional teams with rotating roles.

04 Five-day curriculum

The learning journey: from first agent to final showcase

Students first learn how AI works, then build an agent, test it with users, attack it, fix it and present it.



DAY 1 · BUILD	DAY 2 · INNOVATE	DAY 3 · SECURE	DAY 4 · PRESENT	DAY 5 · PRESENT
"You know the tools. Now build the machine."	*"Ship something real. Learn by building."*	*"Don't just build it. Defend it."*	*"Show the work. Own the learning."*	*"Celebrate the work. Capture the learning."*
AI 101, agentic patterns, first agent, cloud deployment	Design sprint, MVP build, cross-house user testing	Red-team, blue-team fix sprint, governance documents	Cross-house demos, expert Q&A, 1,000-word reflection	Retrospective, final showcase, portfolio, certificates
DELIVERS Working agent pipeline, deployment script, architecture blueprint	**DELIVERS Minimum viable product, user-testing report, iteration log**	**DELIVERS Security audit log, fix commits, release checklist**	**DELIVERS Live demo, peer feedback, reflection essay**	**DELIVERS Portfolio package, cohort insights**

Weekly sequence Build (Day 1) → Innovate (Day 2) → Secure (Day 3) → Present (Days 4–5). Teaching moves from story to concept to lab, so students always see why before how.

05 Day-by-day agenda

What happens in the room, and what every team produces

DAY 1 — BUILD

"You know the tools. Now build the machine."

WHAT STUDENTS LEARN AND DO

- AI 101: rules, symbolic AI, machine learning, generative and multimodal models
- Agentic patterns: ReAct, plan-and-execute, reflection and multi-agent
- Tool calling and retrieval-augmented generation (RAG)
- Build a first agent with LangGraph and a local model
- Deploy it through Docker to the cloud

DELIVERABLES — **Working agent pipeline** ◆ **Deployment script** ◆ **Architecture blueprint**

DAY 2 — INNOVATE

"Ship something real. Learn by building."

WHAT STUDENTS LEARN AND DO

- Design sprint with Crazy 8s sketching, dot voting and storyboards
- Select a client brief and define what the agent should and should not do
- Build in two blocks
- Structured cross-house user testing

DELIVERABLES — **Minimum viable product** ◆ **User-testing report** ◆ **Iteration log**

DAY 3 — SECURE

"Don't just build it. Defend it."

WHAT STUDENTS LEARN AND DO

- Threat landscape: prompt injection, jailbreaks, data exfiltration, PII leakage and supply-chain attacks
- Red-team each house's own product and one other
- Blue-team fix sprint
- Model card, risk assessment, DPIA checklist and human-in-the-loop design

DELIVERABLES — **Security audit log** ◆ **Fix commits** ◆ **Release checklist**

DAY 4
PRESENT

"Show the work. Own the learning."

WHAT STUDENTS LEARN AND DO

- Cross-house demo rotation with peer scoring
- Questions from the facilitator and two external guests
- Written reflection of 1,000 words
- Closing ceremony

DELIVERABLES **Live demo ◆ Peer feedback ◆ Reflection essay**

DAY 5
PRESENT

"Celebrate the work. Capture the learning."

WHAT STUDENTS LEARN AND DO

- Cohort retrospective
- Final showcase covering problem, build, secure and demo
- Portfolio assembly
- Certificates and alumni networking

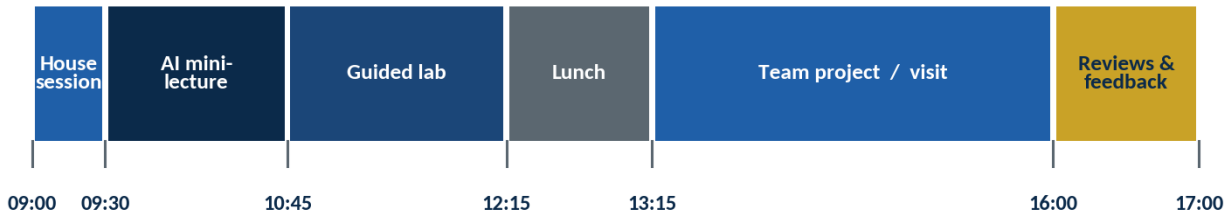
DELIVERABLES **Portfolio package ◆ Cohort insights**

THE DAILY RHYTHM

Every day runs on the same predictable timetable

A DAY AT THE ACADEMY

Story → concept → lab · a question break every ~15 minutes



TIME	SESSION	WHAT HAPPENS
09:00	**House session**	Team check-in, goals for the day and role rotation
09:30	**AI mini-lecture**	Story first, concept second, with a short question break roughly every 15 minutes
10:45	**Guided lab**	Hands-on practice with code and product on screen
12:15 – 13:15	**Lunch**	Break
13:15	**Team project or visit**	House teams build; optional partner visits where the timetable allows
16:00	**Reviews and feedback**	Daily guided feedback for every student

06 The final project standard

What every house team must demonstrate by the end of the week

Agent backbone
Multi-step reasoning, tool use and memory

Evaluation harness
A test set that measures how well the agent performs

Deployed and green
A passing CI/CD pipeline and a live, deployed app

User-validated feature
At least one feature proven through real user testing

Red-team findings, fixed
Attacks found, logged and resolved before release

Governance documents
Model card, risk assessment and release checklist

Recorded live demo
A demonstration students present and defend to experts

"Real AI, really built."
Students leave with a deployed product, not slides.

WHAT STUDENTS TAKE HOME

A working AI product	Security audit log and release checklist	Reflection essay and live demo	Portfolio package of code and video	Certificate and alumni network

07 The house system

Four houses. Six students each. One shared goal.

Students work in four houses inspired by a British school house tradition. Each house is a team of six that rotates roles every day, so everyone practises every part of building an AI product.



GRYFFINDOR
House A
CORE ROLES
Architect and Security Lead



SLYTHERIN
House B
CORE ROLES
Engineer and Innovation Lead



HUFFLEPUFF
House C
CORE ROLES
Data/ML Lead and DevOps Lead



RAVENCLAW
House D
CORE ROLES
Research Lead and Product Designer

House names are working titles. Original names and crests are recommended for public marketing, subject to rights clearance.

EVERY HOUSE TEAM OF SIX



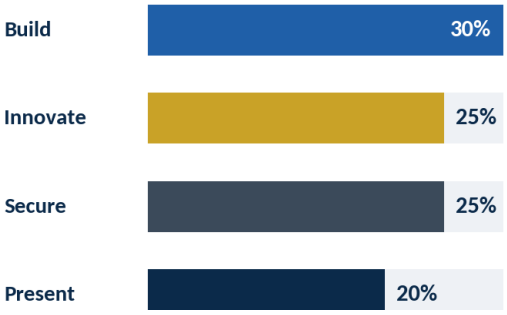
2 Technical | 2 Product & design | 1 Research | 1 Operations

Role rotation Each day students rotate across Architecture, Engineering, Data/ML, DevOps, Security and Product. Each house elects a House Head to coordinate the team.

08 Assessment and recognition

Four pillars, scored for feedback and recognition

ASSESSMENT WEIGHTING



BUILD Architecture quality, completeness, deployment readiness, code quality

INNOVATE Problem solving, user validation, design quality, feature completeness

SECURE Threat identification, fixes, governance compliance, monitoring

PRESENT Demo quality, technical explanation, reflection depth, peer feedback



Recognition: awards, ribbons, house trophies and individual certificates (no cash prizes). Judging panel: the facilitator plus two external guests, such as an industry practitioner, an ethics researcher or a venture investor.

AI ACADEMY FOR THE NEXT GENERATION

"You know the tools. Now build the machine."

Build · Innovate · Secure · Present